

Polymer electrolytes based on the doped comb-branched copolymers for Li-ion batteries

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ARTICLE INFO

Article history:

Received 16 January 2010

Received in revised form 24 May 2010

Accepted 26 May 2010

Keywords:

Poly(4-vinylbenzeneboronic acid)

Polymer electrolyte

Li-ion conductivity

ABSTRACT

In this work, poly(4-vinylbenzeneboronic acid), PVBBA was synthesized via free-radical polymerization of 4-vinylbenzeneboronic acid (4-VBBA) and followed by modification with polyethylene glycol monomethylether (PEGME) with different molecular weights to produce boron containing comb-branched copolymers and abbreviated as PVBBAPEGME_X. Then copolymer electrolytes were successfully prepared by doping of the host matrix with CF₃SO₃Li at several stoichiometric ratios to get PVBBAPEGME_X-Y (X is the molecular weight of PEGME and Y is the EO/Li ratio). The materials were characterized by using Fourier transform infrared spectroscopy (FT-IR), nuclear magnetic resonance (¹H-NMR and ¹¹B-NMR), thermogravimetry (TG) and differential scanning calorimeter (DSC). The ionic conductivity of these novel copolymer electrolytes was studied by dielectric-impedance spectroscopy. Li-ion conductivity of these copolymer electrolytes depends on the length of the side units as well as the doping ratio. Such electrolytes possess satisfactory ambient temperature ionic conductivity (exceeding 10^{−4} S/cm) and good mechanical strength. At higher doping ratios, Vogel–Tamman–Fulcher (VTF) behavior of conductivity implies the coupling of the charge carriers with the segmental motion of the polymer chains.

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1. Introduction

Lithium ion conducting polymer electrolytes have both technological and scientific interest due to their potential applications in solid state lithium ion batteries [1]. These materials should have high ionic conductivity, electrochemical stability and good mechanical properties for application [2]. Generally, solid polymer electrolytes can be prepared by dissolving salt (LiX) in a suitable polymer matrix where there is an interaction between functional units of the host polymer and guest molecules [3–5]. In this context, polyethylene oxide (PEO) based electrolytes are the earliest and the most extensively studied systems [6]. In 1973, the first measurements on conductivities of PEO complexed with alkali metal salts were made by Wright et al. [7]. It was after Armand that the potential of these new materials were realized for future battery applications [8]. Depending on the salt concentration PEO based polymer electrolytes have semi-crystalline morphology which may negatively affect the ionic conductivity especially at lower temperatures. In dry state these dry polymer electrolyte systems (PEO–LiX) offer lower ionic conductivity of the order of 10^{−6} S/cm at ambient temperature.

In addition gel polymer electrolytes can also be prepared by combination of a stable polymer matrix and special additives such as ethylene carbonate (EC), propylene carbonate (PC), dimethyl carbonate

(DMC) and N-methylpyrrolidinone (NMP) (BL) to promote ionic conductivity [9–11]. Polymers such as polyvinylidene fluoride (PVDF), polyacrylonitrile (PAN), poly(methyl methacrylate) (PMMA) and polyvinyl chloride (PVC) have been studied in polymer electrolytes [12–14]. These systems comprise functional groups that are not sufficient to create charge separation between electrolyte ions. Thus, special additives are required to promote sufficient charge separation, which enable the ions in the solid matrix to respond to an electric field. In these systems, the major role of the polymer is mostly to maintain a solid and stable matrix, whereas the ion migration within the matrix under an electrical field is feasible because of the additives. However, the mechanical property of such polymer electrolytes is generally poor because of the characteristic of gels and additives [15].

Moreover, a variety of comb-branched polymers was investigated, in which PEO side chains were attached to a polymer backbone, which itself had a low glass transition temperature and thus giving rise to a more flexible system [16].

Polysiloxanes which include PEO side chains were studied by several groups [17–19]. These systems are liquids at RT and illustrated ionic conductivities of the around 10^{−4} S/cm after mixing with LiClO₄. However, their chemical stabilities are poor due to the Si–O–C branched side chains.

In addition to polysiloxanes polymers, comb-branched systems based on polyphosphazene (MEEP) backbones and PEO side chains were investigated. These materials are completely amorphous with glass transition temperatures around −80 °C. However, the room-temperature conductivities of the F₃CSO₃Li salt mixtures were close to 10^{−5} S/cm [20].

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Poly(4-vinylbenzeneboronic acid), PVBBA is a thermally and chemically stable polymer and used as host matrix for proton conducting membrane [21]. It is an attractive polymeric support for polymer-supported esterification reaction with diols [22]. The functionalization can easily be done by the attachment of hydroxyl groups in a suitable solvent.

In this work, poly(4-vinylbenzeneboronic acid), PVBBA was synthesized via free-radical polymerization of 4-vinylbenzeneboronic acid (4-VBBA) and followed by modification with polyethylene glycol monomethylether (PEGME) with different molecular weights to produce boron containing comb-branched copolymers. The attachment of PEGs to the polymer was investigated by FT-IR, ^1H -NMR and ^{11}B -NMR. Then copolymer electrolytes were produced by doping the comb-branched copolymers with $\text{CF}_3\text{SO}_3\text{Li}$ with several EO/Li ratios. The copolymer electrolytes were characterized via FT-IR, TGA and DSC. Li-ion conducting properties of the materials were investigated by dielectric-impedance analyzer and the results are discussed and compared with previously reported systems.

2. Experimental

2.1. Materials

4-Vinylbenzeneboronic acid (>98%) was provided from Alfa Aesar and the initiator α,α' -azodiisobutyramidine dihydrochloride (AIBADC) was supplied from Aldrich. Poly(ethylene glycol monomethyl ether) (PEGME, average $M_n = 350, 550$ and 750) were purchased from Alfa Aesar Chemical Company. N,N -dimethylformamide (DMF, analytical reagent) and dimethyl sulfoxide (DMSO, analytical reagent) were obtained from Merck. PEGME and the solvents (DMF, DMSO and THF) were dried by using molecular sieves. $\text{CF}_3\text{SO}_3\text{Li}$ was received from Aldrich and stored in the glove box.

2.2. Sample preparation

Poly(4-vinylbenzeneboronic acid), PVBBA, was produced by free-radical polymerization of 4-vinylbenzeneboronic acid in DMF using AIBADC as initiator according to our previous work [21]. The poly-

merization solution was purged with nitrogen and the reaction was performed in DMF at 85°C for 4 h. After polymerization, the resulting solid was filtered and washed several times with excess DMF and then dried under vacuum at 50°C (Fig. 1A).

A mixture of PVBBA and PEGME (stoichiometric molar ratio = 1:2.5) was refluxed in glove box at 100°C and 48 h (Fig. 1B). Then polymer gels were washed in excess diethylether, at least three times, to remove unreacted PEGME. Then the materials were dried under vacuum at 60°C to get comb-branched copolymers with three different PEGME side chains, i.e., PVBBAPEGME350, PVBBAPEGME550 and PVBBAPEGME750.

The copolymer electrolytes were prepared by doping of appropriate amounts of comb-branched copolymers with $\text{CF}_3\text{SO}_3\text{Li}$ salt. The copolymers and salt were dissolved in DMSO together in a flask and stirred for 3 h in nitrogen-filled glove box. The concentration of Li salt was adjusted so that the molar ratio of lithium atoms to ether oxygen atoms in the total PEGME of PVBBAPEGME350, PVBBAPEGME550 and PVBBAPEGME750 was as indicated in Table 1. The samples were abbreviated as PVBBAPEGME X -Y where X denotes the molecular weight of the PEGME and Y is the doping ratio, i.e., EO/Li ($Y = 10, 20, 40$ and 60). The resulting homogeneous viscous solution was poured on a Teflon plate to get thin films of the polymer electrolytes. By heating the solution on the plate at 60°C , polymer electrolyte films with thicknesses ranging from 0.5 to 1 mm were obtained.

2.3. Characterizations

^1H NMR and ^{13}C NMR measurements were performed with Bruker Avance 400 MHz in d-DMSO. ^{11}B -NMR measurement was made in the same solvent using $\text{BF}_3 \cdot (\text{C}_2\text{H}_5)_2\text{O}$ as an external reference.

For FT-IR measurements, thin films of the samples were deposited on a silicon wafer, dried under vacuum and stored in glove box. The IR spectra (4000 – 400 cm^{-1} , resolution 4 cm^{-1}) were recorded with a Perkin Elmer FT-IR spectrum BX.

Thermal stabilities of the polymer electrolytes were examined by TG analyses with a Perkin Elmer STA 6000 Thermal Analyzer. The samples ($\sim 10\text{ mg}$) were heated from room temperature to 700°C under N_2 atmosphere at a scanning rate of $10^\circ\text{C}/\text{min}$.

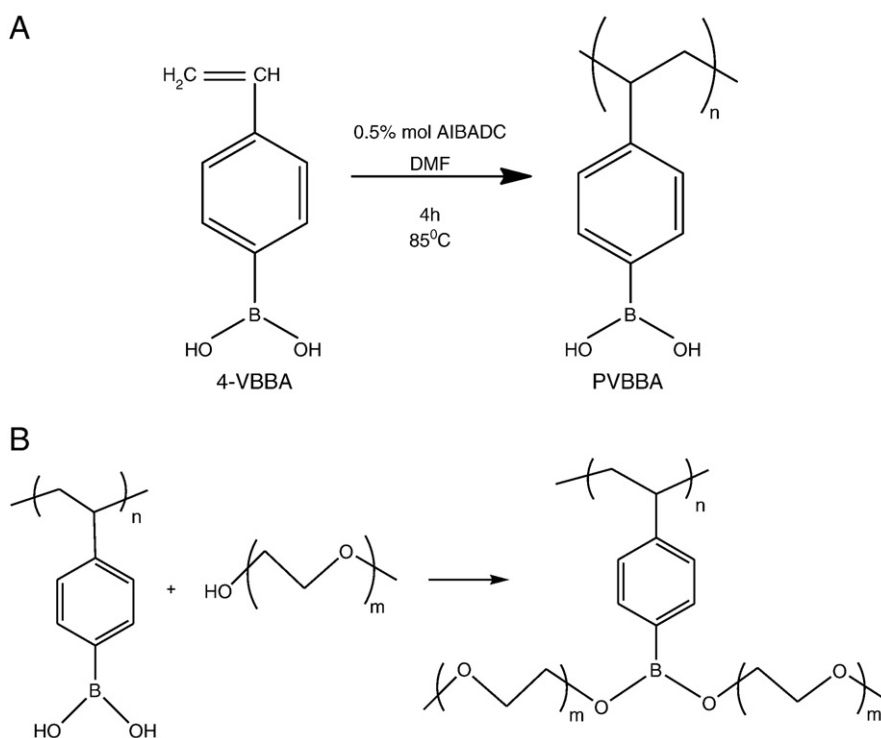


Fig. 1. (A) Synthesis scheme of PVBBA and (B) synthesis scheme of comb-branched copolymers.

Table 1

The composition and properties of PVBBAPEGME-X-Y samples.

Sample PVBBAPEGME-X-Y	X (Mn PEGME)	Y (EO:Li mol ratio)	T _m (°C)	Max. ion conductivity (S/cm)
PVBBAPEGME350	350	–	–10	–
PVBBAPEGME350-10	350	10:1	0	3.5×10^{-3} at 100 °C
PVBBAPEGME350-20	350	20:1	0	1.1×10^{-3} at 100 °C
PVBBAPEGME350-40	350	40:1	0	3.4×10^{-3} at 100 °C
PVBBAPEGME350-60	350	60:1	5	3.7×10^{-8} at 90 °C
PVBBAPEGME550	550	–	–5	–
PVBBAPEGME550-10	550	10:1	–5	1×10^{-3} at 100 °C
PVBBAPEGME550-20	550	20:1	–5	7.7×10^{-4} at 100 °C
PVBBAPEGME550-40	550	40:1	–3	1×10^{-6} at 90 °C
PVBBAPEGME550-60	550	60:1	–2	3×10^{-8} at 100 °C
PVBBAPEGME750	750	–	–	–
PVBBAPEGME750-20	750	20:1	–	7.9×10^{-4} at 100 °C
PVBBAPEGME750-40	750	40:1	–	4.1×10^{-4} at 100 °C
PVBBAPEGME750-60	750	60:1	–	4×10^{-8} at 100 °C

DSC measurements were carried out on a Perkin Elmer Jade DSC under nitrogen atmosphere and heating–cooling curves were recorded at a rate of 10 °C/min. The melt temperature (T_m) was evaluated from the onset temperature of melting. The heat of fusion (ΔH_m) is the enthalpy of fusion of the melting endotherm.

The real (ϵ') and imaginary (ϵ'') parts of complex dielectric permittivity ($\epsilon^* = \epsilon' - i\epsilon''$) were measured with a Novocontrol dielectric-impedance analyzer. The films were sandwiched between platinum blocking electrodes and the conductivities were measured in the frequency range 0.1 Hz to 3 MHz at 10 °C intervals. The temperature was controlled with a Novocontrol Cryosystem, which is applicable between –100 °C and 250 °C. The dielectric data (ϵ' , ϵ'') were collected during heating as a function of frequency. For conductivity data analysis, the real part of conductivity (σ') was calculated from the complex permittivity using the relation, $\sigma'(\omega) = \sigma_{ac}(\omega) = \epsilon''(\omega)\omega\epsilon_0$, where ω is the angular frequency and ϵ_0 is the vacuum permittivity.

3. Results and discussion

4-Vinyl benzeneboronic acid (VBBA) is a boronic acid comprising monomer which becomes highly brittle after polymerization. To improve its mechanical properties a mixture of PEGME and PVBBA was heated until water as a by-product was eliminated. Then well characterized comb-branched copolymers which include different length PEGs as side chains were produced. Then copolymer electrolytes

were synthesized by doping of these novel host matrices with CF₃SO₃Li at several stoichiometric ratios with respect to ethylene oxide (EO) units.

3.1. ¹H-NMR study

¹H NMR spectrum of the synthesized boron containing comb-branched copolymer is shown in Fig. 2. The peaks at 1–3 ppm were assigned to methylene backbone protons (CH₂ and CH) where the C–H near 2.5 ppm peak is overlapped by solvent peak (DMSO). The methoxy protons linked to boronic acid (B–OCH₃) was found at 4.7 ppm. The peaks appearing at 3–4 ppm belong to ethylene oxide protons. By comparison of the integrals of protons corresponding to ethylene oxide and benzyl ring protons almost all the B–OH units successfully reacted with PEGME.

In our previous work, the pure PVBBA, three and four coordinated boron sites were clearly presented. It was reported that four coordinated boron sites (ring C–BO₃) usually appear close to ~0 ppm and three coordinated boron sites (ring C–BO₂) appear at higher chemical shift values. The shift differences are due to major structural features, like the coordination of the boron sites [21]. ¹¹B-NMR result demonstrated that the resonance covering the chemical shift range from –5 ppm up to 5 ppm, is due to three coordinated boron site. The results demonstrated that the grafting occurs predominantly via coordination.

3.2. FT-IR study

The FT-IR spectra of homopolymer, PVBBA is shown in Fig. 3. The OH stretching of –B(OH)₂ group gives a broad peak at around 3380 cm^{–1}. The aromatic CH stretching is located at 3030 cm^{–1} and the absorption bands at 2930 and 2850 cm^{–1} are assigned to the symmetric and anti-symmetric stretching vibration of aliphatic CH units. Two strong peaks at 1660 cm^{–1} and 1606 cm^{–1} are attributed to C=C stretching bands of p-disubstituted benzene ring. The strong CH out-of-plane deformation band for p-disubstituted benzene ring is at 830 cm^{–1}. The vibration of phenyl boronic acid linkage (Ph–B) gives a strong peak at 1415 cm^{–1}. The absorption bands at 1342 and 1105 cm^{–1} are assigned to the vibration of the bond B–O and B–OH. The OH deformation of –B(OH)₂ group is located at 1006 cm^{–1} [23–24].

Fig. 3B shows the FT-IR spectra of comb-branched copolymer, PVBAPEGME750 and doped samples, PVBAPEGME750-20, PVBAPEGME750-40 and PVBAPEGME750-60. The broad absorption band at 3380 cm^{–1} that correspond to absorption of –B(OH)₂ groups, was

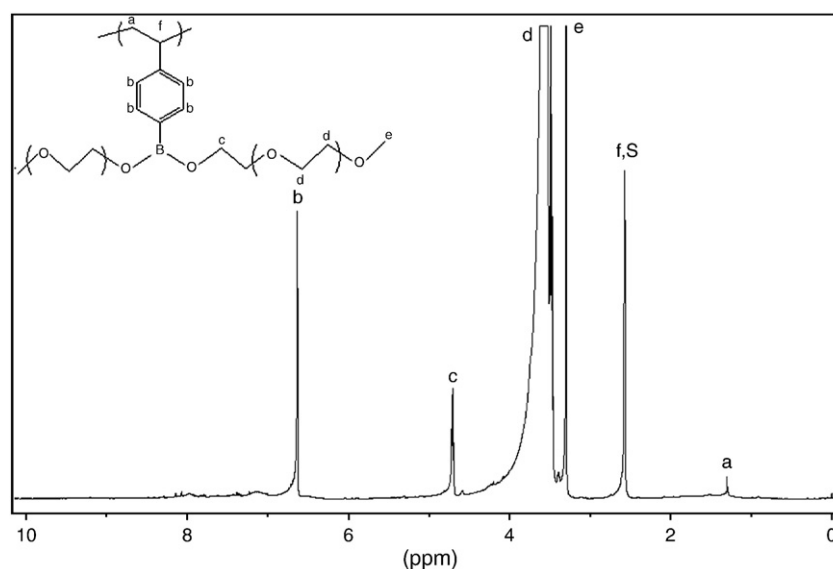


Fig. 2. ¹H NMR spectra of comb-like copolymer (sample in Fig. 1b).

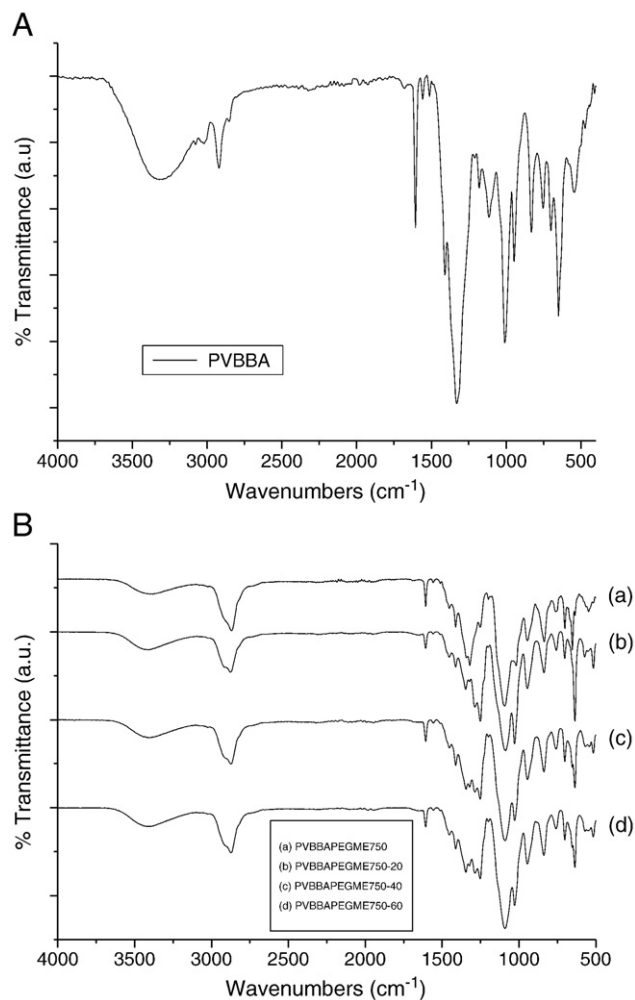


Fig. 3. A. FT-IR spectra of the homopolymer PVBBA. B. FT-IR spectra of PVBBAPEGME750 and the doped samples PVBBAPEGME750-Y.

very weak, which the illustrated high conversion of the hydroxyl groups has confirmed by NMR spectroscopy. The absorption bands near 1343 cm^{-1} were assigned to the anti-symmetric stretching vibration (B–O). The peak at 1030 cm^{-1} can be attributed to stretching vibration of SO_3 unit of dopant. The presence of Li^+ cations in the electrolyte, and its coordination with the ether groups of the polymer, was shown by the significant increase in the intensity of the characteristic absorption band of the C–O–C group at 1100 cm^{-1} [25].

3.3. Thermal analysis

Thermal stability is an important property for polymer electrolytes during application in lithium ion polymer batteries. The thermal stability was studied by TGA for the PVBBA, comb-branched copolymers and the Li salt doped copolymers. Fig. 4 shows the TGA traces of PVBBA, PVBBAPEGME350 and the polymer electrolytes, PVBBAPEGME350-Y. For P4VBBA, two steps of weight loss were noticed where the exponential weight decay within $100\text{--}400^\circ\text{C}$ can be attributed to condensation of B–OH units and then the polymer decomposes. Clearly, this behavior disappeared for PVBBAPEGME350 and no weight change was observed up to at least 250°C . This is additional confirmation of the reaction of B–OH with PEGME units. Similarly, for all the polymer electrolytes PVBBAPEGME350-X, no separate steps of weight loss were observed until 250°C . Fig. 4B shows the thermograms of PVBBAPEGME550 and the polymer electrolytes, PVBBAPEGME550-Y. There is an elusive weight change up to 300°C ($<5\%$) which may be attributed to

evaporation of absorbed humidity or removal of the condensation products as well as the solvent (DMSO). The second step of the weight loss which started above 350°C is due to degradation of the materials. TGA results of the PVBBAPEGME750 and the corresponding polymer electrolytes are illustrated in Fig. 4C. The onset temperature of weight

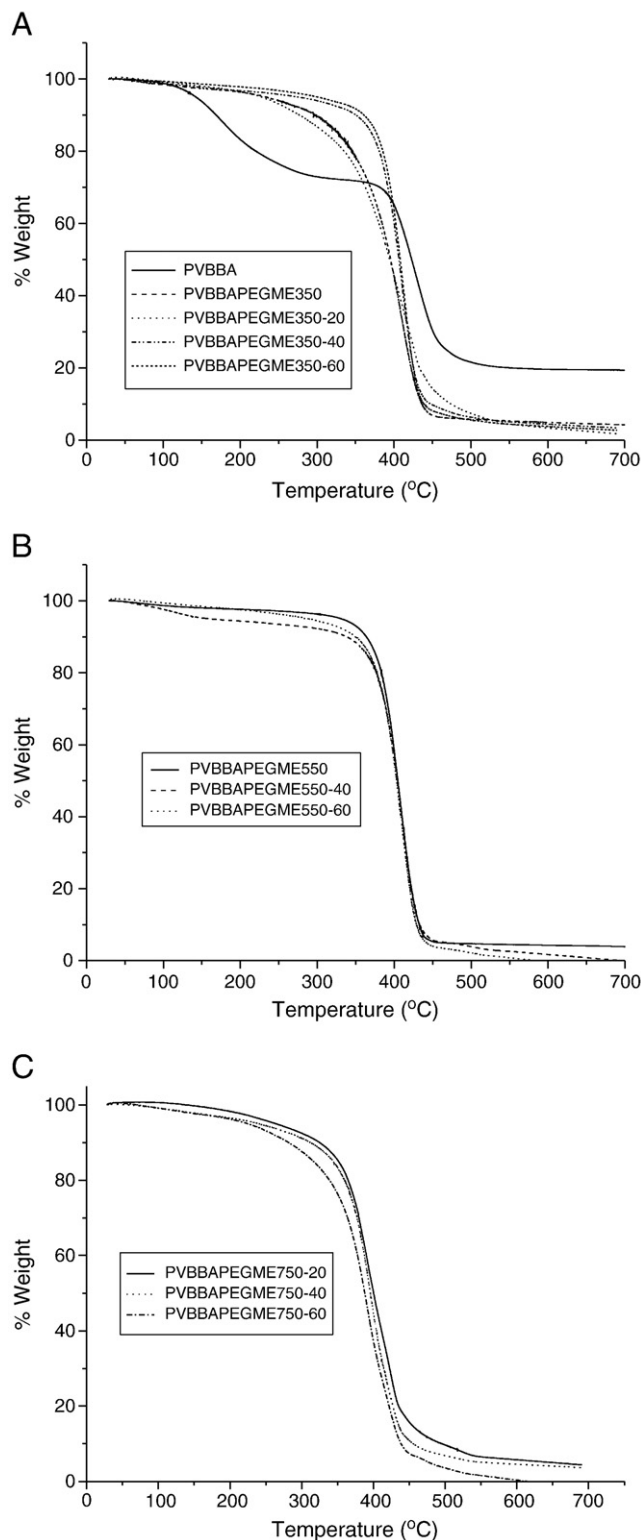


Fig. 4. A. Comparison of the weight loss of PVBBA, PVBBAPEGME350 and the copolymer electrolytes. The heating rate was $10^\circ\text{C}/\text{min}$. B. Comparison of the weight loss of PVBBAPEGME550 and the copolymer electrolytes. The heating rate was $10^\circ\text{C}/\text{min}$. C. Comparison of the weight loss of copolymer electrolytes, PVBBAPEGME750-Y. The heating rate was $10^\circ\text{C}/\text{min}$.

loss is approximately 200 °C. A slight weight change between 200 and 300 °C can be attributed to complex rearrangement of the molecules which is followed by a drastic degradation of the polymer above 300 °C.

Fig. 5A and B shows the melt endotherms of semi-crystalline PEGME of comb-branched copolymer and the doped samples. The copolymer samples PVBBPEGME350 and PVBBAPEGME550 in the absence of dopant illustrate melting temperatures (onset) -10 °C and -5 °C, respectively. The samples PVBBPEGME350-40 and PVBBPEGME350-60 were found to display slightly higher T_m and distinctly higher enthalpy of fusion (ΔH_m) (Fig. 4). Similarly, the melting temperatures of PVBBPEGME550-20, PVBBPEGME550-40 and PVBBPEGME550-60 are -5 °C, -3 °C and -2 °C, respectively. A different trend was observed for PVBBPEGME750-X samples where there was no melting endotherm in the DSC thermograms which indicated that all the samples are amorphous.

3.4. Conductivity measurement

The ion conductive behavior of the copolymer electrolytes with different PEGME pendant side units was studied with impedance spectroscopy. The samples were prepared in the glove box under nitrogen atmosphere with a humidity <0.1 ppm. Also the conductivity

measurement was carried out in a completely water-free environment so that the contribution of humidity to the ion conductivity was prevented. The frequency and temperature dependent AC conductivities (σ_{ac}) of PVBBAPEGME350-10 and PVBBAPEGME550-10 are shown in Fig. 6A and B. The σ_{ac} versus frequency curve change with the frequency where in the low frequency region, the reduction in the conductivity is due to polarization of the blocking electrodes. For example in Fig. 6, at low temperature and high frequency ($F < 10^3$ Hz, at 20 °C) the conductivity increase is attributed to dispersions in polymer.

The DC conductivity (σ_{dc}) of the samples was obtained from the plateaus of $\log \sigma_{ac}$ versus $\log F$ by linear fitting of the data (i.e., $2 \times 10^{-3} - 3 \times 10^6$ Hz, at 40 °C and $2 \times 10^{-4} - 3 \times 10^6$ Hz, at 70 °C).

The lithium ion conductivity can be related to copolymer structure and it is reasonable to draw a model for lithium salt in this host matrix system. In this context, the copolymer has boron containing stiff PVBBA chains as reinforcing backbone and flexible side chains (PEGME) as the matrix for ion transport (Fig. 7). In the novel comb-like copolymer system, lithium trifluoromethane sulfonate salt can be easily dissociated into cations and anions, which are dispersed and stabilized by oxygen atoms and methylene groups in the copolymer chains, respectively. When a potential is applied, cations tend to move towards the cathode while anions move in the opposite direction. At the same time, the

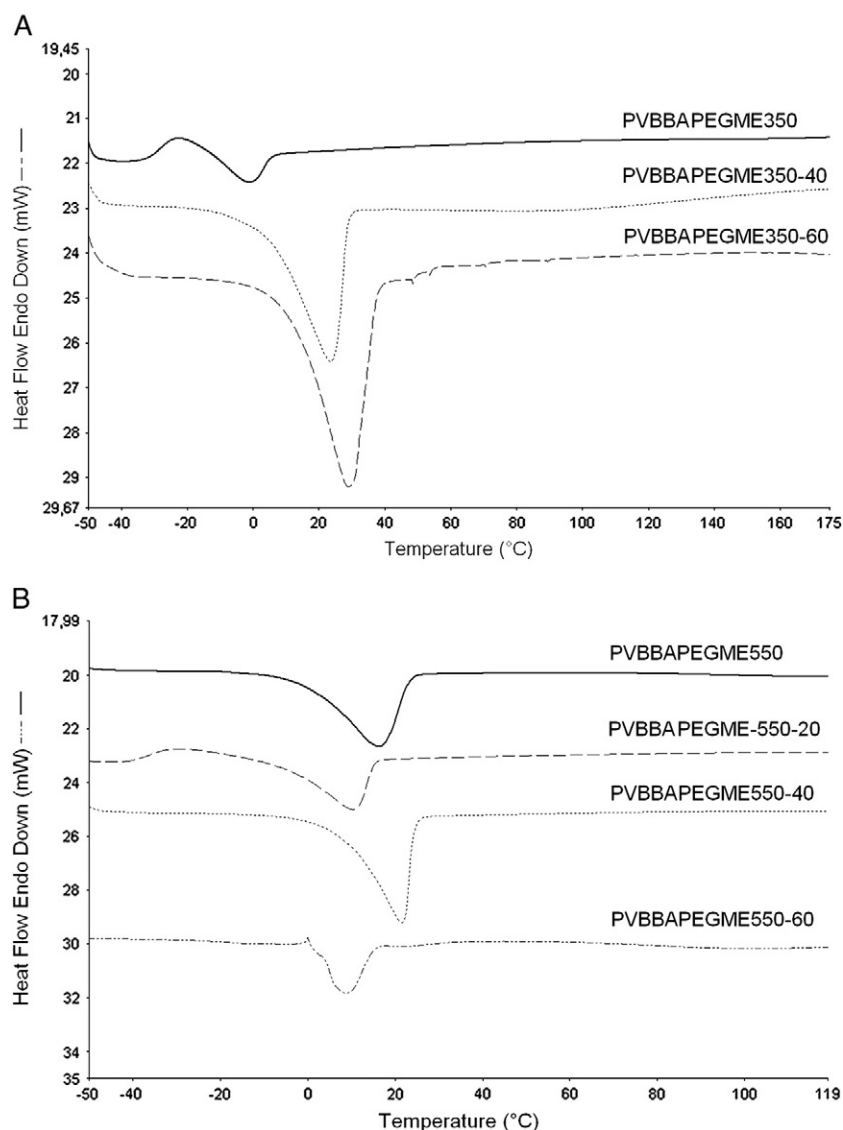


Fig. 5. A. DSC curves of PVBBAPEGME350 and the copolymer electrolytes. Second heating run endotherms are presented at a heating rate of 10 °C/min. B. DSC curves of PVBBAPEGME550 and the copolymer electrolytes. Second heating run endotherms are presented at a heating rate of 10 °C/min.

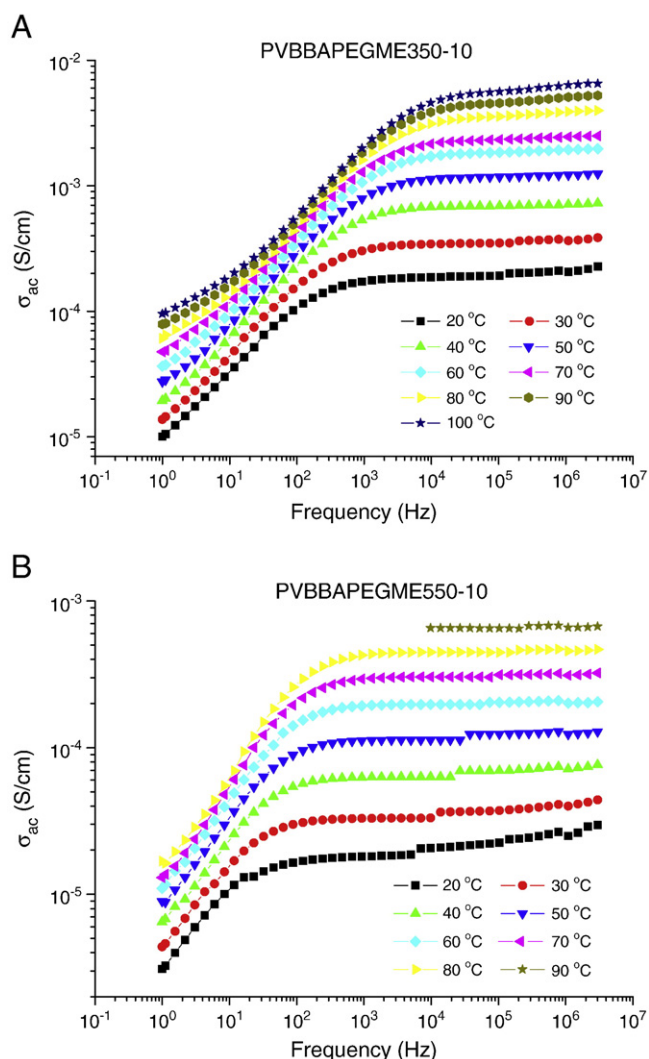


Fig. 6. A. AC conductivity versus frequency of anhydrous PVBBAPEGME350-10 at various temperatures. B. AC conductivity versus frequency of anhydrous PVBBAPEGME550-10 at various temperatures.

charges on the interface of electrode and electrolyte will rearrange. In Fig. 8 the temperature dependence of ionic conductivities for all the copolymer/salt complexes are compared in the Arrhenius coordinates. As seen, the ionic conduction mechanism of the polymer electrolytes PVBBAPEGME $\times$ -Y (Y is in the range 10–40) follows Vogel–Tamman–Fulcher (VTF) equation (Eq. (1))

$$\sigma = \sigma_0 \exp -B / k(T - T_0) \quad (1)$$

where σ_0 is a pre-exponential factor which may be related to ion conductivity at infinite temperature, B is an apparent activation energy, k is Boltzmann constant, and T_0 is the temperature which corresponds to onset of the segmental relaxations. The curved conductivity isotherms correspond to those polymer electrolytes that have higher ionic conductivity and shorter side chains. It seems that for these amorphous comb-like copolymers the flexibility of polymer side chains partially determines the ionic conductivity.

For these non-plasticized copolymer electrolytes the measured conductivities are exceptionally high where it is 4×10^{-3} S/cm at 100 °C and 1.6×10^{-4} S/cm at 20 °C. Such an ionic conductivity can be described by molecular structure of the host polymer. Clearly, faster lithium ion conduction occurs in the polymer matrix by the fast conduction pathways created by highly ordered arrangement of Li^+

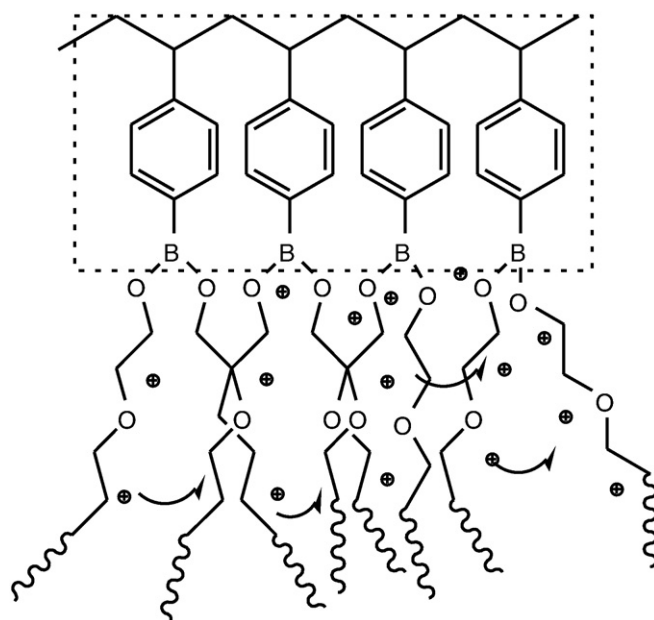


Fig. 7. Model of the structure of PVBBAPEGME $\times$ -Y.

coordination EO sites along the PVBBA backbone. This situation is schematically illustrated in Fig. 7.

The incorporation of boron as Lewis acid in ethylene oxide based structures was previously reported by Pennarun et al. [25]. The boron was expected to increase the dissociation of lithium salt between the electrolyte and the device due to its affinity to both the anion and the ether oxygen [26]. The Li salt anion was assumed to make ionic coordination with boron so accumulation and density of anionic charge at the surface of the counter electrode was expected to be reduced by Lewis acid during application [27].

Previously Lauter et al. used poly(p-phenylene) (PPP) as the rigid backbone with flexible oligo(ethylene oxide) (EO) side chains attached, which form the liquid matrix in which Li salts can be dissolved and in which ionic transport takes place [28]. The materials were named PPP (EO) $\times$ /y with x/y indicating the number of ethylene oxide units in the side chains of a copolymer. When the matrix was plasticized by the addition of TEGDME the ionic conductivity increased and become comparable to our system, PVBBAPEGME $\times$ -Y, which we report in this work.

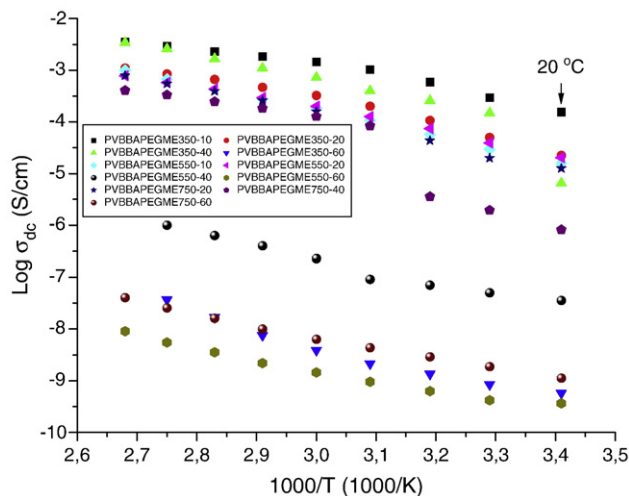


Fig. 8. Temperature dependence of the DC conductivities of PVBBAPEGME $\times$ -Y.

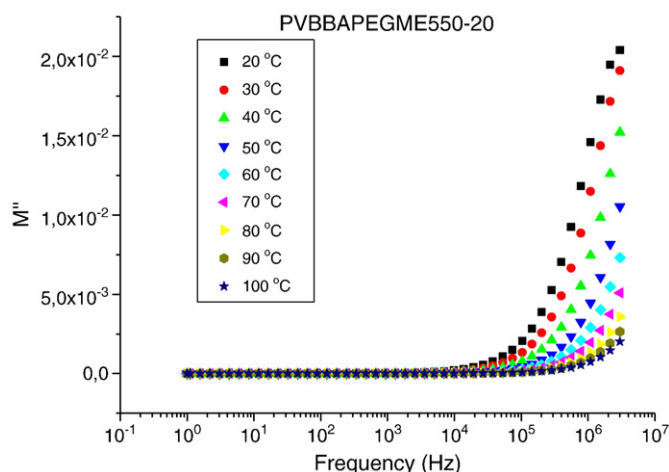


Fig. 9. Variation of imaginary part of modulus (M'') as a function of frequency for PVBBA-PEGME550-20 at different temperatures.

Surprisingly, in the absence of plasticizer like material, films were prepared from PVBBAPEGME550-20 which exhibit surprisingly good mechanical properties. This appears to be the result of the reinforcing effect of PVBBA backbones comprising hairy PEGME side chains.

3.5. Electrical modulus

The frequency dependence of M'' at different temperatures is shown in Fig. 9. It is seen from the figure that at lower frequencies M'' approaches to zero indicating negligible contribution of electrode polarization. At higher frequencies, M'' increases gradually with a tendency for saturation. The presence of the peaks in the modulus formalism at higher frequencies implies that the copolymer electrolyte films are ionic conductors. The observed high frequency dispersion is mainly due to conductivity relaxation. This peak position shifts to higher frequencies with increasing temperature.

4. Conclusions

In the present work, a novel comb-branched boron containing copolymers with different PEGME side chains were produced. Then copolymer electrolytes were synthesized after doping the host material with lithium salt at several molar ratios, i.e., EO/Li. The branch EO units of the copolymer is capable of forming donor:acceptor type bonds by complexing with the metal ion. These copolymer electrolytes were characterized using DSC, FT-IR, ^1H -NMR and dielectric-impedance spectroscopy. ^{11}B -NMR result demonstrated the three coordinated

boron site. The conductivity studies illustrate that the ionic conductivity of these polymer electrolytes was found to depend on side chain length as well as the Li^+ content. The polymer electrolyte PVBBAPEGME550-20 illustrated a satisfactory ionic conductivity of $1.6 \times 10^{-4} \text{ S/cm}$ at 20°C and $4 \times 10^{-3} \text{ S/cm}$ at 100°C . The polymer electrolytes with higher Li^+ content shows VTF behavior, indicating that the Li^+ ion motion is dependent on the flexibility of the polymer side chain. These materials have good mechanical strength and can be suggested for application to lithium batteries.

Acknowledgements

This work was supported by BOREN under contract number 2009.ÇÖ213.

Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version, at [doi:10.1016/j.ssi.2010.05.039](https://doi.org/10.1016/j.ssi.2010.05.039).

References

- [1] A.M. Spethan, K.S. Nahm, *Polymer* 47 (2006) 5952.
- [2] J.M. Tarascon, M. Armand, *Nature* 414 (2001) 359.
- [3] K.M. Abraham, *Electrochim. Acta* 38 (1993) 1233.
- [4] P.G. Bruce, *Electrochim. Acta* 40 (1995) 2077.
- [5] Z. Gadjeva, Y.G. Andreev, D.P. Tunstall, P.G. Bruce, *Nature* 412 (2001) 520.
- [6] P. Lightfoot, M.A. Metha, P.G. Bruce, *Science* 262 (1993) 883.
- [7] D.E. Fenton, J.M. Parker, P.V. Wright, *Polymer* 14 (1973) 589.
- [8] M.B. Armand, S.M. Chabagno, M. Duclot, Second International Meeting on Solid Electrolytes, St. Andrews, Scotland, Extended Abstracts, 1978, 20–22 Sept.
- [9] Y.K. Yarovoy, H.P. Wang, S.L. Wunder, *Solid State Ionics* 118 (1999) 301.
- [10] H.P. Wang, H. Huang, S.L. Wunder, *J. Electrochem. Soc.* 147 (2000) 2853.
- [11] K. Murata, S. Izuchi, Y. Yoshihisa, *Electrochim. Acta* 45 (2000) 1501.
- [12] Y.J. Shen, M.J. Reddy, P.P. Chu, *Solid State Ionics* 175 (2004) 747.
- [13] Q. Xiao, X. Wang, W. Li, Z. Li, T. Zhang, H. Zhang, *J. Membr. Sci.* 334 (2009) 117.
- [14] D.J. Lin, C.L. Chang, C.K. Lee, L.P. Cheng, *Eur. Polym. J.* 42 (2006) 2407.
- [15] M. Wang, L. Qi, F. Zhao, S. Dong, *J. Power Sources* 139 (2005) 223.
- [16] W.H. Meyer, *Adv. Mater.* 10 (1998) 439.
- [17] P.G. Hall, G.R. Davis, J.E. McIntyre, I.M. Ward, D.J. Banister, K.M.F. Le Brocq, *Polym. Commun.* 27 (1986) 98.
- [18] D. Fish, I.M. Khan, J. Smid, *Macromol. Chem. Rapid Commun.* 7 (1986) 115.
- [19] R. Spindler, D.F. Shriver, *J. Am. Chem. Soc.* 10 (1988) 3036.
- [20] P.M. Blonsky, D.F. Shriver, P. Austin, H.R. Allcock, *J. Am. Chem. Soc.* 106 (1984) 6854.
- [21] A. Sezgün, Ü. Akbey, M.R. Hansen, R. Graf, A. Bozkurt, A. Baykal, *Polymer* 49 (2008) 3859.
- [22] G. Belogi, T. Zhu, G.J. Boons, *Tetrahedron Lett.* 41 (2000) 6965.
- [23] G. Kahraman, O. Beşkardeş, Z.M.O. Rzaev, E. Pişkin, *Polymer* 45 (2004) 5813.
- [24] M. Şenel, A. Bozkurt, A. Baykal, *Ionics* 13 (2007) 263.
- [25] P.Y. Pennarun, P. Jannasch, S. Papaefthymiou, N. Skarpentzos, P. Yianoulis, *Thin Solid Films* 514 (2006) 258.
- [26] F. Zhou, D.R. MacFarlane, M. Forsyth, *Electrochim. Acta* 48 (2003) 1749.
- [27] S.K. Mohapatra, *J. Electrochem. Soc.* 125 (1978) 284.
- [28] U. Lauter, W.H. Meyer, G. Wegner, *Macromolecules* 30 (1997) 2092.